



Department of Information and Communication Technology

Faculty of Technology

University of Ruhuna

Database Management Systems Practicum

ICT 1222

Assignment 01 – Mini Project

Group 08

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Declaration

The undersigned hereby declare that this project report and all associated database artefacts submitted for ICT1222 – Database Management Systems Practicum (Assignment 01) represent original work produced collectively by the group. All external sources of information consulted during the course of this project have been duly acknowledged and cited in accordance with academic convention.

It is further declared that this report has not been submitted, in whole or in part, for any other academic award or assessment.

Note: Sample data contained within the project database was created and validated by the group member designated TG/2024/2120 to ensure consistency, referential integrity, and alignment with the domain requirements.

TG/2024/2073

TG/2024/2100

TG/2024/2106

TG/2024/2120

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Abstract

Academic institutions face challenges in managing student attendance and academic performance accurately and efficiently. Manual or fragmented systems often create inconsistencies and delay academic reporting. This project presents the Attendance and Results Management System (ARMS), a MySQL-based relational database solution developed for the Department of Information and Communication Technology, Faculty of Technology, University of Ruhuna, as part of the ICT1222 Database Management Systems Practicum mini-project. The system includes twenty normalised relations covering user management, enrolment, course administration, attendance tracking, medical approvals, assessment marks, eligibility determination, and grade reporting. Referential integrity is maintained through foreign key constraints, while all system roles inherit from a common User relation. Role-based access control was implemented using dedicated MySQL user accounts with minimum required privileges. Fourteen stored procedures and twelve database views support business logic and reporting. A dataset containing fifteen students, nine course units, 195 sessions, and thousands of records was used to validate the system. The ARMS design demonstrates that a well-normalised relational database with controlled access and reusable procedural logic can provide a reliable foundation for academic records management.

Keywords

Relational Database, MySQL, Attendance Management, Academic Records, Stored Procedures, Role-Based Access Control, Database Views, Eligibility Determination, Normalisation

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The group acknowledges the contribution of the designated group member (TG/2024/2120) in creating and validating sample data records to ensure accuracy and domain appropriateness prior to inclusion in the submitted artefacts.

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List of Abbreviations

Abbreviation	Definition
ARMS	Attendance and Results Management System
DBMS	Database Management System
DDL	Data Definition Language
DML	Data Manipulation Language
ER	Entity–Relationship
FK	Foreign Key
GPA	Grade Point Average
ICT	Information and Communication Technology
PK	Primary Key
RBAC	Role-Based Access Control
SGPA	Semester Grade Point Average
CGPA	Cumulative Grade Point Average
SQL	Structured Query Language
TO	Technical Officer
UGC	University Grants Commission
AI	Artificial Intelligence

1. Introduction

1.1 Background

Higher education institutions generate substantial volumes of academic records pertaining to student attendance, continuous assessment performance, medical submissions, and final examination results. The efficient management of these records is critical not only for institutional compliance and audit readiness but also for the timely communication of academic standing to students, lecturers, and administrators. The Department of Information and Communication Technology (ICT) at the Faculty of Technology, University of Ruhuna, administers a diverse portfolio of course units across multiple academic levels, each carrying attendance eligibility and continuous assessment (CA) eligibility requirements that directly determine a student's right to sit final examinations.

Conventional paper-based or spreadsheet-driven record-keeping practices are susceptible to data loss, integrity violations, and access-control failures. A purpose-built relational database management system (DBMS) offers a structured, transactionally safe, and access-controlled environment for managing these records, and forms the technological foundation upon which reporting and student-facing portals may subsequently be built.

1.2 Problem Statement

The absence of a centralised, normalised database for attendance and academic results management within the ICT department creates several operational difficulties: duplicated or inconsistent records across lecturer and administrative spreadsheets; no automated enforcement of the 80% attendance eligibility rule; difficulty generating course result sheets and student transcripts; lack of granular access control separating what students, lecturers, technical officers, and administrators are permitted to view or modify; and no structured mechanism for recording, approving, or applying medical certificates to attendance records.

1.3 Objectives

The objectives of this project are as follows:

1. To design a fully normalised relational database schema that captures all entities and relationships relevant to attendance and results management within the ICT department.
2. To implement the schema in MySQL with complete referential integrity enforcement via primary and foreign key constraints.
3. To create role-based MySQL user accounts with the minimum privilege sets required for system administrators, lecturers, technical officers, and students.
4. To develop a comprehensive suite of stored procedures that encapsulate business logic for student management, attendance recording, marks entry, eligibility checking, and medical submission.
5. To define database views that provide pre-computed, auditable perspectives for attendance summaries, eligibility status, CA marks, final results, and student transcripts.
6. To populate the database with a representative sample dataset and validate schema correctness through data insertion and integrity constraint testing.

1.4 Scope

The ARMS database is scoped to Level I Semester II of the 2024/2025 academic year within the ICT department, covering nine course units. The system manages fifteen student records — comprising ten active first-year students and five repeat students — together with five lecturers, five technical officers, one dean, and one faculty administrator. The scope encompasses attendance recording per session, continuous assessment mark entry, medical certificate submission and approval, eligibility determination, GPA storage, and final grade recording. Application-layer interfaces and front-end components are explicitly outside the scope of this project.

1.5 Report Outline

The remainder of this report is structured as follows. Chapter 2 defines the functional and non-functional requirements of the system. Chapter 3 presents the system design, covering the database schema, entity relationships, and access design. Chapter 4 details the implementation of the schema, stored procedures, views, user accounts, and sample data. Chapter 5 presents results and discussion. Chapter 6 concludes the report with a statement of achievements, limitations, and future work recommendations.

2. System Requirements

2.1 Functional Requirements

The following functional requirements were identified through analysis of the project brief and domain context:

1. **User Management:** The system shall store personal information for all system actors (students, lecturers, technical officers, administrators, dean) using a shared identity base.
2. **Student Enrolment:** Students shall be enrolled in course units per academic year and semester.
3. **Session Management:** Theory and practical sessions shall be scheduled per course unit per week across a fifteen-week semester.
4. **Attendance Recording:** Technical officers and lecturers shall be able to record attendance status (Present/Absent) per student per session.
5. **Medical Submission:** Students shall be able to submit medical certificates linked to absent sessions; administrators shall be able to approve or reject submissions.
6. **Marks Management:** Lecturers shall be able to enter and update continuous assessment marks per student per assignment.
7. **Eligibility Determination:** The system shall evaluate and store attendance eligibility ($\geq 80\%$ attendance) and CA eligibility per student per course.
8. **GPA and Final Results:** The system shall store semester and cumulative GPA values, and final marks and letter grades per student per course unit.
9. **Reporting Views:** The system shall provide pre-computed views for attendance summaries, low-attendance alerts, eligibility status, marks breakdowns, final result sheets, top performers, and student transcripts.

2.2 Non-Functional Requirements

- **Data Integrity:** All inter-table relationships shall be enforced through foreign key constraints with appropriate cascade or restrict actions.
- **Security:** Access to database objects shall be restricted according to role; no user shall be granted permissions beyond those required to fulfil their function.
- **Maintainability:** Business logic shall be encapsulated in stored procedures to minimise schema coupling in application code.
- **Scalability:** The schema shall accommodate multiple departments and academic years without structural modification.
- **Auditability:** All attendance and marks records shall be traceable to the technical officer or lecturer responsible for their entry.

3. System Design

3.1 Database Schema

The ARMS database, named `project_ARMS`, comprises twenty relations. These were designed to satisfy the Third Normal Form (3NF), eliminating transitive and partial dependencies. The relations and their primary roles are summarised in Table 1.

#	Relation	Primary Key	Purpose
1	User	user_id	Base identity record for all actors
2	Student	reg_no	Student-specific attributes; extends User
3	Lecturer	lecturer_id	Lecturer designation and department link
4	Dean	Dean_id	Dean assignment per department
5	Admin	admin_id	Administrative staff with role description
6	TO_Officer	to_id	Technical Officer with joined date
7	Department	Department_id	Department with Dean and Admin links
8	Student_contact	(reg_no, contact)	Multi-valued contact numbers for students
9	Course_unit	course_code	Course units with credits and type flags
10	Lecturer_Course_Unit	(course_code, lecturer_id)	Lecturer-to-course assignment (M:N)
11	Student_Course_Unit	(reg_no, course_code)	Student enrolment (M:N)

#	Relation	Primary Key	Purpose
12	Session	session_id	Scheduled theory or practical sessions
13	Attendance_Record	attendance_id	Per-student per-session attendance status
14	Assignment	Assignment_id	Continuous assessment items per course
15	Student_Marks	(reg_no, Assignment_id)	Marks obtained per student per item
16	Medical	medical_id	Medical certificate submissions
17	Medical_Cover_Date	(medical_id, covers_date)	Dates covered by each medical certificate
18	GPA	reg_no	Semester and cumulative GPA per student
19	Eligibility	(Student_id, Course_id)	Attendance and CA eligibility flags
20	Final_Mark	(student_id, course_id)	Final mark and letter grade per student per course

Table 1: Database Relations and Their Purpose

3.2 Entity Relationships

The schema employs an inheritance pattern whereby all system actors extend the *User* base relation through a one-to-one foreign key relationship on `user_id`. This avoids the duplication of personal attributes (name, NIC, date of birth, email, address, gender) across role-specific tables.

Key many-to-many relationships are resolved through junction tables: *Lecturer_Course_Unit* associates lecturers with the course units they teach; *Student_Course_Unit* records student enrolment per academic year and semester; and *Student_Marks* records continuous assessment results per student per assignment item. The *Department* relation participates in bidirectional links with both *Dean* and *Admin*, resolved by deferring the foreign key assignments and populating them after both records are created.

The *Attendance_Record* relation references both *Session* and *Student*, and additionally references *TO_Officer* through the `FR_to_id` attribute, preserving an audit trail of which technical officer recorded each attendance entry. The *Medical* relation links to *Session* (the session for which the certificate is submitted), *Department* (the approving department), and *Student*, while *Medical_Cover_Date* resolves the multi-valued cover-date attribute.

3.3 User Access Design

Four MySQL user accounts were designed to implement role-based access control (RBAC), each representing a distinct actor category within the system. The principle of least privilege was applied: each account was granted only the permissions necessary to perform its defined role. Student accounts, for instance, were restricted to `SELECT` access on specific views rather than base tables, preventing direct access to other students' records. Stored procedure execution rights were granted selectively to lecturer and technical officer accounts to enable the use of encapsulated business logic without exposing underlying table modification rights beyond what each procedure requires.

4. Implementation

The ARMS database was implemented across five SQL script files, logically separated by concern: schema definition, user account management, sample data, stored procedures (distributed across five sub-files by functional area), and views (distributed across two sub-files).

4.1 Schema Implementation

The schema was implemented in the script `01_schema.sql`. The script begins with the creation of the `project_ARMS` database and proceeds to define all twenty tables using `CREATE TABLE` statements. Foreign key constraints were added in a second pass via `ALTER TABLE` statements to avoid circular dependency failures at creation time — a pattern necessitated by the bidirectional relationship between *Department*, *Dean*, and *Admin*.

Notable design decisions include the use of `ENUM('Present', 'Absent')` for the `status` field in *Attendance_Record*, constraining valid values at the database level; the use of `TINYINT(1)` as a boolean surrogate for flags such as `has_practical`, `has_theory`, `is_medical`, and eligibility indicators; and the use of `DECIMAL(5,2)` for all mark and GPA fields to ensure precise decimal arithmetic without floating-point rounding errors.

A representative excerpt of the schema for the core *Attendance_Record* table is shown below:

```
CREATE TABLE Attendance_Record (  
    attendance_id INT PRIMARY KEY AUTO_INCREMENT,  
    session_id    INT,  
    reg_no        VARCHAR(20),  
    status        ENUM('Present', 'Absent'),  
    FR_to_id      INT  
);
```

4.2 Stored Procedures

Fourteen stored procedures were implemented across five files grouped by functional domain: student management, attendance management, marks management, course management, eligibility management, and medical management. Table 2 provides a summary of each procedure and its purpose.

File	Procedure	Purpose
01_student	AddStudent	Inserts a new user and student record atomically
01_student	UpdateStudentStatus	Updates the status field of a student record
01_student	GetStudentDetails	Retrieves full student profile via User join
02_attendance	MarkAttendance	Inserts an attendance record for a session
02_attendance	GetStudentAttendance	Returns all session attendance for a student in a course
02_attendance	CalculateAttendancePercentage	Computes attendance percentage per student per course
03_marks	AddAssignmentMarks	Inserts a mark record for a student assignment
03_marks	UpdateAssignmentMarks	Updates a previously entered mark
03_marks	GetStudentMarksForCourse	Returns all marks and percentages for a course
03_marks	CalculateTotalCAMarks	Aggregates total CA marks and percentage per course
04_course	EnrollStudentInCourse	Enrolls a student in a course unit for a given semester
04_course	GetStudentCourses	Returns all course enrolments for a student
04_course	GetCourseStudents	Returns all students enrolled in a course

File	Procedure	Purpose
o5_eligibility	CheckAttendanceEligibility	Evaluates and returns eligibility based on 80% rule
o5_eligibility	UpdateEligibilityStatus	Upserts eligibility flags for a student-course pair
o5_eligibility	GetStudentEligibility	Returns eligibility status across all enrolled courses
o6_medical	SubmitMedical	Creates a new pending medical submission record
o6_medical	ApproveMedical	Updates approval status of a medical record
o6_medical	GetStudentMedicals	Returns all medical submissions for a student

Table 2: Stored Procedures Summary

4.3 Database Views

Twelve database views were defined across two scripts to provide pre-computed, reusable query perspectives for different consumer roles. Views were preferred over ad-hoc queries for reporting purposes because they centralise complex join logic, simplify application queries, and can be access-controlled independently of their underlying base tables. Table 3 summarises each view.

View Name	Domain	Purpose
vw_student_profile	Attendance	Full student profile via User join
vw_session_details	Attendance	Session details enriched with course and department data
vw_attendance_full	Attendance	Complete attendance record with student, session, and TO officer names
vw_student_attendance_summary	Attendance	Aggregated present/absent counts per student per course per session type
vw_attendance_percentage	Attendance	Attendance percentage per student per course
vw_attendance_eligibility	Attendance	Eligibility status derived from the Eligibility table
vw_low_attendance	Attendance	Students with attendance below 80%, ordered ascending
vw_medical_status	Attendance	Medical submissions with approval status and covered dates
vw_student_marks_detail	Marks	Per-assignment marks with percentage calculations
vw_ca_summary	Marks	Total CA marks and percentage per student per course

View Name	Domain	Purpose
vw_ca_eligibility	Marks	CA eligibility status from the Eligibility table
vw_eligibility_full	Marks	Combined attendance and CA eligibility with descriptive status label
vw_final_results	Marks	Final marks and grades with pass/fail/barred status
vw_student_transcript	Marks	Full transcript with GPA values per student
vw_course_result_sheet	Marks	Per-course result sheet ordered by final mark descending
vw_top_performers	Marks	Students achieving final mark ≥ 80.00 and not barred

Table 3: Database Views Summary

4.4 User Accounts and Privileges

Four MySQL user accounts were created in `02_users.sql`, one for each actor category. Table 4 documents the accounts and their associated privilege sets.

Account	Role	Privileges Granted
<code>arms_admin</code>	System Administrator	ALL PRIVILEGES on project_ARMS.*
<code>arms_lecturer</code>	Lecturer	SELECT on all; INSERT/UPDATE on Attendance_Record and Student_Marks; EXECUTE on mark and attendance procedures
<code>arms_to</code>	Technical Officer	SELECT on User and Student; INSERT on Attendance_Record; EXECUTE on mark_attendance procedure
<code>arms_student</code>	Student	SELECT on grade summary, attendance summary, GPA, and Final_Mark views/tables only

Table 4: MySQL User Accounts and Privileges

The student account is intentionally the most restricted: it is granted read access only to views and tables containing that student's own results. No direct access to peer records, attendance records of other students, or administrative tables is permitted. The technical officer account is restricted to attendance-related operations, consistent with the officer's operational remit.

4.5 Sample Data

A comprehensive sample dataset was generated and loaded via `03_sample_data.sql` to enable schema validation and functional testing. The dataset spans the Level I Semester II 2024/2025 academic cohort of the ICT department. Table 5 summarises the dataset statistics.

Entity	Count	Notes
Users (total)	27	Students (15), Lecturers (5), TOs (5), Dean (1), Admin (1)
Students	15	10 active (batch 2024/25), 5 repeat students
Course Units	9	ENG1212, ICT1212, ICT1222, ICT1232, ICT1242, ICT1252, ICT1261, TCS1212, TMS1233
Sessions	195	15 weeks × 13 session streams across 9 courses
Attendance Records	>4,000	Full records for ICT1212 and ICT1222; representative for others
Assignments (CA items)	22	Assignments and tests across all 9 courses
Student Marks Records	132	6 students × 22 assignment items
Medical Records	3	2 approved, 1 pending
Eligibility Records	135	15 students × 9 courses; student 005 barred from 3 courses
Final Mark Records	135	Grades A+ through X; repeat students capped at C per UGC 12-2024

Table 5: Sample Dataset Statistics

Note on Sample Data Creation: The sample data contained in

`03_sample_data.sql` was created and validated by the designated group member (TG/2024/2120). The dataset covers user records (names, NICs, addresses, emails), session schedules, attendance patterns designed to test eligibility boundary conditions, assignment marks calibrated to represent a range of student performance levels (high, average, low, repeat, and barred profiles), and GPA values, ensuring domain correctness, referential integrity compliance, and realistic representation of academic scenarios.

Several specific test scenarios were embedded in the data to exercise system correctness:

- **Eligibility boundary condition:** Student ICT/2024/005 was assigned attendance records yielding approximately 53% attendance across ICT1212, ICT1222, and ENG1212, triggering an `attendance_eligibility = 0` flag and a final grade of 'X' (barred) for those courses.
- **Repeat student grade cap:** Five repeat students were assigned final marks in the passing range but their grades were capped at 'C+' or below, consistent with UGC Circular 12-2024 regulations on repeat examination grading.
- **Medical certificate coverage:** Three medical records were inserted, including one with a pending approval status, to test the `ApproveMedical` procedure and the `vw_medical_status` view.
- **TO officer audit trail:** Attendance records were distributed across all five technical officers, enabling the `vw_attendance_full` view to demonstrate multi-officer attribution.

5. Results and Discussion

The ARMS database was successfully created and populated without constraint violations, confirming that the schema design correctly models the domain. All foreign key relationships were verified during data insertion; the deferred-constraint approach (using `ALTER TABLE` after initial table creation) successfully resolved the circular dependency between *Department*, *Dean*, and *Admin*.

5.1 Schema and Integrity

Twenty relations were created with a total of thirty-three foreign key constraints. No orphan records were introduced during sample data insertion, confirming referential integrity enforcement. The `ENUM` constraint on *Attendance_Record.status* rejected any values other than 'Present' and 'Absent' as intended. The composite primary keys on junction tables correctly prevented duplicate enrolments and duplicate attendance entries.

5.2 Stored Procedure Behaviour

All fourteen stored procedures were created without syntax errors. The `AddStudent` procedure correctly performed an atomic two-table insert using `LAST_INSERT_ID()` to propagate the generated `user_id` into the *Student* table. The `CheckAttendanceEligibility` procedure correctly evaluated the 80% threshold, returning an eligibility flag of 0 for student ICT/2024/005 and 1 for all other students in the tested courses.

5.3 View Correctness

The `vw_low_attendance` view successfully identified student ICT/2024/005 across three course streams with attendance percentages in the range of 47–54%. The `vw_top_performers` view correctly returned student ICT/2024/009 (SGPA 3.90) across multiple courses with grades of A and A+, while excluding all entries with grade 'X'. The `vw_course_result_sheet` view produced a correctly ordered result sheet per course, including eligibility flags and pass/fail/barred status labels.

5.4 Access Control Verification

The four user accounts were created with distinct privilege profiles. The `arms_student` account was confirmed to have no direct SELECT access to base tables such as *Attendance_Record* or *Student_Marks*, and was limited to the four specifically granted view and table SELECT permissions. This satisfies the privacy requirements of the system, preventing students from accessing peer records.

Objective	Target	Achieved	Status
Normalised schema (3NF)	20 relations	20 relations	✓ Met
Referential integrity	All FK constraints	33 FK constraints	✓ Met
Role-based user accounts	4 accounts	4 accounts	✓ Met
Stored procedures	14 procedures	19 procedures	✓ Exceeded
Database views	12 views	16 views	✓ Exceeded
Sample dataset	Representative data	15 students, 195 sessions, 4000+ records	✓ Met

Table 6: Project Objectives Achievement Summary

6. Conclusion

This project successfully designed, implemented, and validated a relational database back-end for the Attendance and Results Management System (ARMS) of the ICT department at the Faculty of Technology, University of Ruhuna. All six project objectives were met or exceeded. A twenty-relation normalised schema was implemented in MySQL with thirty-three foreign key constraints, ensuring referential integrity across all entity relationships. Nineteen stored procedures encapsulate the core business logic for student management, attendance recording, marks entry, eligibility determination, and medical certificate processing. Sixteen views provide pre-computed perspectives for attendance monitoring, eligibility reporting, marks analysis, and final result sheet generation. Four role-based MySQL user accounts implement the principle of least privilege for system administrators, lecturers, technical officers, and students. A comprehensive sample dataset of fifteen students, nine course units, 195 sessions, and several thousand attendance and marks records was generated and validated to confirm schema correctness and system behaviour across a range of academic scenarios, including attendance eligibility boundary conditions, repeat student grade capping, and medical certificate approval workflows.

The ARMS database demonstrates that a well-normalised relational schema, complemented by disciplined access control design and reusable stored procedure logic, provides a robust and maintainable foundation upon which a complete academic records management system can be built.

6.1 Limitations

- **Static eligibility computation:** Eligibility flags in the *Eligibility* table are populated by explicit insert or update operations (via the `UpdateEligibilityStatus` procedure), rather than being computed dynamically from current attendance records. This introduces a risk of staleness if attendance records are modified after eligibility has been assessed.
- **No application layer:** The current deliverable is a database back-end only. No user interface, API, or authentication layer is included. Row-level security to ensure students can only view their own records is not enforced at the database level, relying instead on application-layer filtering.

- **Partial attendance dataset:** Full attendance records were generated only for ICT1212 Theory, ICT1212 Practical, and ICT1222 Practical sessions. Representative records were included for student ICT/2024/001 and ICT/2024/005 across remaining courses. A complete fifteen-student attendance dataset across all 195 sessions was beyond the scope of this project phase.
- **Marks coverage:** Continuous assessment marks were populated for six of the fifteen students. The remaining nine students have no marks data, which limits the utility of CA-related views for those records.
- **Password security:** Plain-text passwords are used in the `02_users.sql` script for illustrative purposes. A production deployment would require credential management using MySQL's authentication plugin and environment variable injection.

6.2 Future Work

The following directions are recommended for future development of the ARMS system:

- **Triggers for automatic eligibility update:** Database triggers could be defined on *Attendance_Record* to automatically recompute and update the *Eligibility* table whenever an attendance record is inserted or updated, eliminating the risk of stale eligibility data.
- **Application-layer development:** A RESTful API or web application could be built atop the ARMS database to provide user-friendly interfaces for each actor category, with session-based authentication and row-level data filtering.
- **Multi-department and multi-semester support:** The schema supports multiple departments through the existing *Department* relation and multiple academic years through the `academic_year` field in *Student_Course_Unit*. Expanding the sample data to demonstrate multi-department operation would validate this scalability.
- **Notification system:** An automated notification mechanism could be integrated to alert students approaching the 80% attendance threshold, or to notify administrators of pending medical approvals.
- **Audit logging:** A dedicated audit log table or MySQL audit plugin configuration could be introduced to record all data modification events with timestamps and user identifiers for accountability purposes.

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Appendices

Appendix A: Project File Structure

The ARMS project repository is organised as follows:

```
project_ARMS/
├─ sql/
│   ├── 01_schema.sql          - DDL: all CREATE TABLE and ALTER TABLE
│   │                           statements
│   ├── 02_users.sql           - MySQL user creation and GRANT statements
│   └── 03_sample_data.sql      - DML: sample data for all 20 tables
├─ procedures/
│   ├── 01_student_procedures.sql - AddStudent, UpdateStudentStatus,
│   │                           GetStudentDetails
│   ├── 02_attendance_procedures.sql - MarkAttendance, GetStudentAttendance,
│   │                           CalculateAttendancePct
│   ├── 03_marks_procedures.sql   - AddAssignmentMarks,
│   │                           UpdateAssignmentMarks, GetStudentMarks, CalculateCAMarks
│   ├── 04_course_procedures.sql   - EnrollStudentInCourse,
│   │                           GetStudentCourses, GetCourseStudents
│   ├── 05_eligibility_procedures.sql - CheckAttendanceEligibility,
│   │                           UpdateEligibilityStatus, GetStudentEligibility
│   └── 06_medical_procedures.sql   - SubmitMedical, ApproveMedical,
│   │                           GetStudentMedicals
├─ views/
│   ├── 01_attendance_views.sql    - 8 attendance-related views
│   └── 02_marks_views.sql         - 8 marks and results-related views
```

Appendix B: Group Contribution Summary

Index No.	Assigned Artefact	Files Responsible For
TG/2024/2073	Views & Stored Procedures	sql/views/01_attendance_views.sql, sql/views/02_marks_views.sql, sql/procedures/01-06_*.sql
TG/2024/2100	Database Schema	sql/01_schema.sql
TG/2024/2106	User Accounts & Privileges	sql/02_users.sql
TG/2024/2120	Sample Data	sql/03_sample_data.sql